COMPUTER ENGINEERING

QUEZON CITY

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| **COURSE CODE/TITLE:** | CPE 010 - Data Structure & Algorithms | **SECTION:** | CPE21S4 |
| **DATE PERFORMED:** | Aug 20, 2026 | **DATE SUBMITTED:** | Aug 20, 2026 |
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**Assignment 6.1** **Linear and Binary Search**

## 1. PROCEDURE OUTPUT

What procedures did you perform?

**PROCEDURE:**

**Procedure that performed:**

Answer the following questions:

1. What is a search tree in data structures?

- Binary Search Tree is a hierarchical, node-based data structure designed for efficient data retrieval, insertion, and deletion. Where each node has at most two children, It follows a strict ordering rule for any given node, all values in its left subtree are smaller, and all values in its right subtree are larger. This unique layout allows to skip half of the remaining tree during a search operation, mirroring the efficiency of binary search on an array

2. What are the different types of search algorithms in data structures? Differentiate each type of search.

**Sequential Search:** Traverses elements in a strict linear order from start to finish. It requires no sorting and works on both sorted and unsorted data structures (like arrays or linked lists).

- **Linear Search:** Checks every item in a list one by one, starting from the very first element and moving in sequence until it finds a match. Because it does not care about organization, it is perfect for messy, unsorted data. However, it becomes slow for large lists because it might look at every single item before finishing

**Interval Search:** Operates exclusively on sorted data by jumping or dividing the range into indexed intervals or sub-segments. It discards massive blocks of unpromising data instantly, yielding logarithmic time complexities

- **Binary Search:** Is a much faster method that works exclusively on sorted data. It starts by looking at the exact middle item, compares it to the target, and immediately discards the incorrect half of the list. It then repeats this half-and-half splitting process until it isolates the target, saving

massive amounts of time.

- **Ternary Search:** Is a variation of binary search that cuts a sorted list into three equal parts instead of two. By using two middle markers, it evaluates which of the three blocks contains the target and throws away the remaining two blocks. While it discards data aggressively, it requires more checks per step, making it less common for basic lookups.
- **Jump Search**: This algorithm works on sorted arrays. Instead of checking one by one, it skips ahead by fixed steps (or blocks) to find the zone where the element sits, then does a linear search in that block.
- **Interpolation Search**: This is an upgraded version of Binary Search for sorted, evenly distributed data. It guesses where the item might be (like looking up a word starting with "Z" near the back of a physical dictionary) instead of always checking the exact middle.

3. What operations / implementations can be performed using binary and linear search operations?

- **Linear search** is great for messy lists because it checks items one by one. This simple method lets you find the biggest or smallest numbers, count how many times an item shows up, and find all the spots where a matching item is hiding. It is also perfect for checking if a new item is already a duplicate before you add it, and it easily walks through connected chains of data where you are not allowed to skip around.
- **Binary search** only works on sorted lists and finds things fast by cutting your remaining choices in half at every single step. This method finds the exact first or last time a number repeats, finds the closest matching number, or finds the numbers right above or below your target. It can also quickly count repeating items, find the highest peak in a list that goes up and then down, find items in lists that have been shifted out of order.

4. What are the advantages in using binary search trees as data structure?

- A Binary Search Tree is a way to organize data so you can find, add, or delete items much faster than using a simple list. It works like a family tree where smaller numbers go to the left and bigger numbers go to the right, which lets you skip past half the numbers at every single step. Unlike a rigid array (has a fixed size), this tree can grow or shrink automatically whenever you want, it never forces you to slide other items out of the way when making changes, and it keeps everything in perfect order from smallest to biggest automatically.

5. Give an example program using binary search and Linear search.

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**ILO A**

**Task Title & Task Objective**

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| **Screenshot of the code** | |
| **Output of the code** | |
| **Analysis** | |

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| **ILO B** |

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| **Task Title & Task Objective** |
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| **Screenshot of the code** | |
| **Output of the code** | |
| **Analysis** | |

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

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| **SUPPLEMENTARY:** |

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| **ILO C** |

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| **Task Title & Objective:** |
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| **Screenshot of the code** | |
| **Output of the code** | |

| Analysis | |
|---|---|

| Conclusion |
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## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** _____

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** _____

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:** _____

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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